

4- FortiGate WAN VLAN Interface Configuration

Overview

This guide demonstrates how to create a VLAN interface on a FortiGate firewall and configure it as a PPPoE WAN interface when the Internet service provider requires VLAN tagging.

Lab Environment

Firewall: FortiGate VM64

Parent Interface: WAN (port2)

VLAN Interface: VLAN35_WAN

VLAN ID: 35

Addressing Mode: PPPoE

Related Video: <https://www.youtube.com/watch?v=mzsQg1lu3L8>

Step 1 - Navigate to Interfaces

Navigate to Network > Interfaces and edit the physical interface (port2).

Step 2 - Configure port2 as the WAN Interface

Alias: WAN

Role: WAN

Addressing Mode: Manual

IP Address: 0.0.0.0/0.0.0.0 (no IP address assigned to the parent interface)

Disable the administrative access and LLDP options that are not required, then select OK. The physical interface is now ready to act as the parent interface for the WAN VLAN.

The screenshot shows the FortiGate GUI's 'Edit Interface' window for 'WAN (port2)'. The left sidebar shows the navigation menu with 'Network' selected. The main panel displays the following configuration:

- Name:** WAN (port2)
- Alias:** WAN
- Type:** Physical Interface
- VRF ID:** 0
- Role:** WAN
- Estimated bandwidth:** 0 kbps Upstream, 0 kbps Downstream
- Addressing mode:** Manual (selected), DHCP, PPPoE
- IP/Netmask:** 0.0.0.0/0.0.0.0
- Secondary IP address:** Disabled
- Administrative Access:**
 - IPv4: ☐ HTTPS, ☐ SSH, ☐ RADIUS Accounting, ☐ PING, ☐ SNMP, ☐ Security Fabric Connection
 - IPv6: ☐ FMG-Access, ☐ FTM, ☐ Speed Test
- Receive LLDP:** Use VDOM Setting, Enable, **Disable** (selected)
- Transmit LLDP:** Use VDOM Setting, Enable, **Disable** (selected)
- Traffic Shaping:**
 - Outbound shaping profile: Disabled
 - Outbound bandwidth: Disabled

At the bottom, there are 'OK' and 'Cancel' buttons. The FortiGate logo and version 'v7.4.3' are visible in the bottom left corner.

Step 3 - Create the VLAN Interface

Navigate to Network > Interfaces and select Create New > Interface.

Step 4 - Configure the VLAN Interface

Name: VLAN35_WAN

Alias: VLAN35

Type: VLAN

Interface: WAN (port2)

VLAN ID: 35

Role: WAN

Step 5 - Configure PPPoE

Select PPPoE as the addressing mode and enter the username and password provided by the ISP.

Step 6 - Configure the Default Gateway

Enable “retrieve default gateway from server”.

The Unnumbered IP can remain 0.0.0.0 unless the ISP provides a specific value.

Step 7 - Configure Administrative Access

Enable only the administrative access services required in your environment.

Best Practice: Allow administrative access from trusted networks only.

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Dashboard > Edit Interface

Network >

Policy & Objects >

Security Profiles >

VPN >

User & Authentication >

System >

Security Fabric >

Log & Report >

Name: VLAN35 (VLAN WAN)

Alias: VLAN35

Type: VLAN

VLAN protocol: 802.1Q

Interface: WAN (port2)

VLAN ID: 35 [Edit](#)

VRF ID: 0

Role: WAN

Estimated bandwidth: 0 kbps Upstream, 0 kbps Downstream

☐ Dedicated Management Port

Address

Addressing mode: Manual DHCP **PPPoE**

Status

Username: Username

Password: [Change](#)

Unnumbered IP: 0.0.0.0

Initial Disc Timeout: 1

Initial PADT Timeout: 1

Retrieve default gateway from server: ☒

Distance: 5

Override internal DNS: ☐

OK Cancel

FORTINET v7.4.3

Technical Notes

- port2 acts as the physical parent interface for the WAN VLAN.
- The VLAN ID must match the value provided by the ISP.
- PPPoE credentials are provided by the ISP.
- Enable only the administrative access protocols required in your environment.

Result

- The physical port2 interface is configured as a WAN interface, and VLAN35_WAN is created on top of it.
- The VLAN interface is ready to establish PPPoE Internet connectivity through the ISP-provided VLAN.

The screenshot displays the FortiGate VM64-KVM configuration interface. On the left, a sidebar menu lists various configuration sections: Dashboard, Network, Policy & Objects, Security Profiles, VPN, User & Authentication, System, Security Fabric, and Log & Report. The main panel shows the configuration for the WAN interface. At the top, there's a header for 'FortiGate VM64-KVM' with a search bar and a grid of icons. Below this, there are buttons for 'Create New', 'Edit', 'Delete', and 'Integrate Int'. A table lists the interfaces, with 'WAN' selected. The table has columns for 'Name' and 'Type'. The 'WAN' interface is shown as a 'Physical Interface' (WAN (port2)) and a 'VLAN' (VLAN35 (VLAN_WAN)).

Name	Type
Undefined 3	
WAN 2	
WAN (port2)	Physical Interface
VLAN35 (VLAN_WAN)	VLAN